

**HOW TO USE THIS SET**

20 questions, mixed difficulty, 10 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.**EASY**

Which are the three major salivary glands?

- A. Submandibular, sublingual and von Ebner's
- B. Parotid, sublingual and lingual
- C. Parotid, submandibular and labial
- D. Parotid, submandibular and sublingual
- E. Parotid, labial and palatine

2.**EASY**

Which epithelium lines the oral cavity?

- A. Pseudostratified ciliated columnar
- B. Simple columnar
- C. Stratified squamous
- D. Simple cuboidal
- E. Transitional

3.**EASY**

Which lingual papillae contain no taste buds?

- A. Fungiform
- B. Foliate
- C. All four types contain taste buds
- D. Circumvallate
- E. Filiform

4.**EASY**

Which is the largest salivary gland?

- A. Submandibular
- B. Labial gland
- C. Sublingual
- D. Gland of von Ebner
- E. Parotid

5.**MEDIUM****CASE**

A section of tongue shows a papilla surrounded by a groove. Ducts of serous glands open into the groove and taste buds line the lateral wall of the papilla.

Which papilla is this, and which taste does it detect?

- A. Filiform; bitter
- B. Circumvallate; sweet
- C. Foliate; salty
- D. Circumvallate; bitter
- E. Fungiform; sweet and salty

6.**MEDIUM**

The glands of von Ebner are:

- A. glands opening beside the lingual frenulum
- B. serous glands opening into the grooves round circumvallate papillae
- C. sebaceous glands of the vermilion border
- D. labial glands opening onto the inner lip
- E. mucous glands in the submucosa of the soft palate

7.

MEDIUM

CASE

A lip section shows lightly keratinized stratified squamous epithelium, prominent blood vessels giving a red colour, numerous sensory nerve endings, and no hair follicles, sweat or sebaceous glands.

Which zone of the lip is this?

- A. Outer skin surface
- B. Vermilion border (transitional zone)
- C. Cheek mucosa
- D. Ventral surface of the tongue
- E. Inner (oral) surface

8.

MEDIUM

What separates the anterior two-thirds of the dorsal tongue from the posterior third?

- A. a V-shaped groove, with lingual tonsils behind it
- B. the palatoglossal arch, with taste buds behind it
- C. the uvula and soft palate
- D. the frenulum, with palatine tonsils behind it
- E. a transverse ridge of filiform papillae

9.

MEDIUM

CASE

In a salivary gland section, round groups of pyramidal cells with basophilic cytoplasm and a narrow central lumen lie next to larger, more acidophilic cells arranged as tubules.

Which pair of structures is shown?

- A. Myoepithelial cells beside interlobular ducts
- B. Intercalated ducts beside striated ducts
- C. Mucous acini beside serous tubules
- D. Two groups of serous acini
- E. Serous acini beside mucous tubules

10.

MEDIUM

Striated ducts are characterised by:

- A. a lining of myoepithelial cells
- B. abundant surrounding connective tissue and stratified epithelium
- C. a narrow lumen and simple cuboidal epithelium
- D. location in the capsule of the gland
- E. basal striations and a simple columnar epithelium

11.

MEDIUM

CASE

A duct lies in a connective tissue septum between two lobules. It is surrounded by abundant connective tissue and its lining varies from simple columnar to stratified epithelium.

Which duct is this?

- A. Duct of a von Ebner gland
- B. Striated duct
- C. Interlobular duct
- D. Intercalated duct
- E. Serous acinus

12.

MEDIUM

CASE

A small tumour compresses a single long duct as it opens into the mouth opposite the second upper molar.

Which gland is drained by this duct?

- A. Submandibular
- B. Glands of von Ebner
- C. Parotid
- D. Sublingual
- E. Labial minor glands

13.

MEDIUM CASE

A gland section shows mainly serous acini with some mucous tubules capped by serous demilunes. Striated ducts are long and intercalated ducts are short.

Which gland is this?

- A. Submandibular
- B. Labial minor gland
- C. Parotid
- D. Gland of von Ebner
- E. Sublingual

14.

MEDIUM

Which statement about the palate is correct?

- A. Soft palate: mucosa on bone; hard palate: muscle core + glands
- B. Both palates: keratinized, with hair follicles
- C. Neither palate has a submucosa or glands
- D. Hard palate: mucosa on bone; soft palate: muscle core + mucous glands
- E. Both palates: skeletal muscle core, no glands

15.

HARD CASE

After an injury to the back of the tongue, a patient can still perceive sweet at the tip but has lost bitter taste.

Damage to which structures explains the loss, and where do the serous gland ducts of these structures open?

- A. Circumvallate papillae; into the grooves surrounding them
- B. Fungiform papillae; onto the tip of the tongue
- C. Foliate papillae; into the sulcus terminalis
- D. Filiform papillae; between the papillae
- E. Circumvallate papillae; into the floor of the mouth beside the frenulum

16.

HARD CASE

Reading a lip section from outside in: skin with hair follicles → a zone of lightly keratinized epithelium with tall dermal papillae and no hair, sweat or sebaceous glands → thick non-keratinized epithelium with small clumps of glands. Deeper, bundles of striated muscle run concentrically round the oral orifice.

Which interpretation is correct?

- A. The middle zone is the inner mucosa; the deep muscle is the tongue musculature; the inner glands are von Ebner's glands
- B. The middle zone is the vermilion border; the deep muscle is orbicularis oris; the glands on the inner surface are labial (minor salivary) glands
- C. The middle zone is the vermilion border; the deep muscle is skeletal but arranged longitudinally; the inner glands are parotid tissue
- D. The middle zone is the vermilion border; the deep muscle is smooth muscle; the inner glands are sweat glands
- E. The middle zone is the outer skin; the deep muscle is orbicularis oris; the inner glands are sebaceous only

17.

HARD CASE

Three gland sections are identified. A: purely serous acini, long intercalated and short striated ducts. B: serous-predominant mixed acini with serous demilunes, long striated ducts. C: mucous tubules with serous demilunes, drained by 10–12 small ducts.

Which ranking of saliva output, from greatest to least, is correct?

- A. C > A > B
- B. B > A > C
- C. A > C > B
- D. A > B > C
- E. B > C > A

18.

HARD

CASE

A pathologist traces secretion from a parotid acinus outwards, from the smallest duct within a lobule to the larger ducts between lobules.

Which is the correct path from the acinus to the septa?

- A. Interlobular → striated → intercalated duct
- B. Striated → interlobular → intercalated duct
- C. Striated → intercalated → interlobular duct
- D. Intercalated → striated → interlobular duct
- E. Intercalated → interlobular → striated duct

19.

HARD

Which set correctly matches each gland with its duct opening and share of saliva?

- A. Parotid: opposite the 2nd upper molar, ~25-30%; submandibular: floor of mouth beside the frenulum, ~70%; sublingual: 10-12 small ducts, ~5%
- B. Parotid: floor of mouth beside the frenulum, ~70%; submandibular: opposite the 2nd upper molar, ~25-30%; sublingual: 10-12 small ducts, ~5%
- C. Parotid: opposite the 2nd upper molar, ~5%; submandibular: 10-12 small ducts, ~70%; sublingual: beside the frenulum, ~25-30%
- D. Parotid: 10-12 small ducts, ~25-30%; submandibular: opposite the 2nd upper molar, ~70%; sublingual: beside the frenulum, ~5%
- E. Parotid: opposite the 2nd upper molar, ~70%; submandibular: 10-12 small ducts, ~25-30%; sublingual: beside the frenulum, ~5%

20.

HARD

Which statement about the tongue is correct?

- A. It is smooth muscle bound loosely to its mucosa by submucosa
- B. A thick fatty layer separates the mucosa from the muscle
- C. Muscle bundles cross in three planes; the ventral surface is non-keratinized
- D. The dorsal surface lacks papillae; the ventral surface bears them
- E. Muscle bundles run in one plane; the ventral surface is heavily keratinized

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	D	The major glands are the parotid, submandibular and sublingual; the minor glands are numerous small glands in the mouth.
2	C	The oral mucous membrane is stratified squamous epithelium resting on connective tissue.
3	E	Filiform papillae give roughness and friction but carry no taste buds.
4	E	The parotid is the largest salivary gland.
5	D	A papilla in a groove with von Ebner gland ducts and lateral taste buds is circumvallate = bitter.
6	B	Excretory ducts of serous glands open into the grooves surrounding circumvallate papillae (glands of von Ebner).
7	B	Lightly keratinized, red from prominent vessels, many sensory endings, no hair / sweat / sebaceous glands = vermillion border.
8	A	A V-shaped groove (sulcus terminalis) divides the anterior $\frac{2}{3}$ (oral cavity) from the posterior $\frac{1}{3}$ (oropharynx); lingual tonsils lie behind it.
9	E	Serous cells are pyramidal, basophilic and form spherical acini with a narrow lumen; mucous cells are larger, more acidophilic and tubular.
10	E	Striated ducts show basal striations with simple columnar cells; intercalated ducts are narrow and cuboidal.
11	C	Interlobular ducts run in the septa, are larger than intralobular ducts, sit in abundant CT, and vary from simple columnar to stratified.
12	C	One long parotid duct opens into the oral cavity opposite the second upper molar on each side.
13	A	Mixed serous + mucous with serous predominance, serous demilunes and long striated / short intercalated ducts = submandibular.
14	D	Hard palate: mucosa on bone. Soft palate: skeletal muscle core + numerous mucous glands in the submucosa.
15	A	Bitter is detected by circumvallate papillae at the back of the tongue; von Ebner's serous glands open into their surrounding grooves.
16	B	Vermilion = transitional zone without adnexa; deep skeletal muscle (orbicularis oris) is concentric around the orifice; inner surface carries labial glands.
17	B	B = submandibular (~70%), A = parotid (~25-30%), C = sublingual (~5%).
18	D	Intralobular ducts (intercalated then striated) drain into the larger interlobular ducts located in the septa.
19	A	Parotid ~25-30% (2nd upper molar); submandibular ~70% (either side of frenulum); sublingual ~5% (10-12 small ducts).
20	C	Tongue = skeletal muscle bundles in 3 planes bound tightly to the mucosa; ventral epithelium is non-keratinized, dorsal is partly keratinized with papillae.

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 1, Dr Hero K. Mustafa
Written from the lecture slides only – check any answer you doubt against your lecture notes.